

Autoencoder Analysis Report

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1 Task Observations

1.1 Reconstruction Quality across Latent Sizes

The three models converged to noticeably different reconstruction losses.

Latent Dimension	Final Train Loss	Final Val Loss
2	0.0256	0.0254
32	0.0105	0.0107
64	0.0105	0.0108

Table 1: Final MSE losses for each model after 30 epochs.

The 2-dimensional model is worse, roughly $2.4\times$ higher loss than the 32 and 64 models. The 32 and 64 models converge to nearly identical losses, suggesting that the bottleneck shifts from the latent dimension to the MLP capacity around 32 dimensions.

1.2 Latent Space Visualisation

- T-shirt and Shirt are close to each other reflecting similarity.
- Sneaker and Ankle boot also occupy the similar region of the plot.
- Bag is relatively isolated and is distinctive.
- Trouser is well separated from upper-body garments.

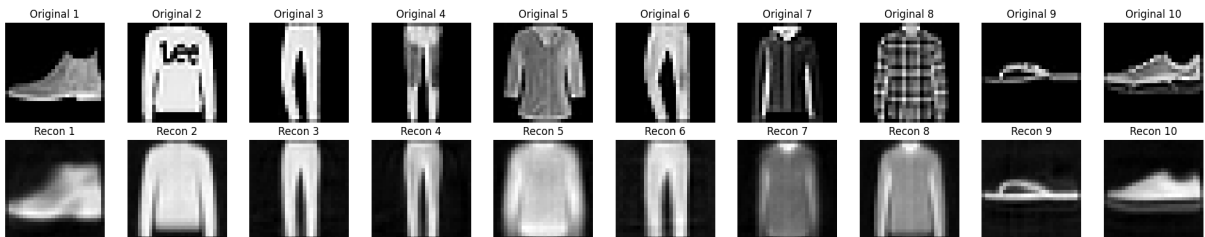


Figure 1: 2 dimension reconstruction

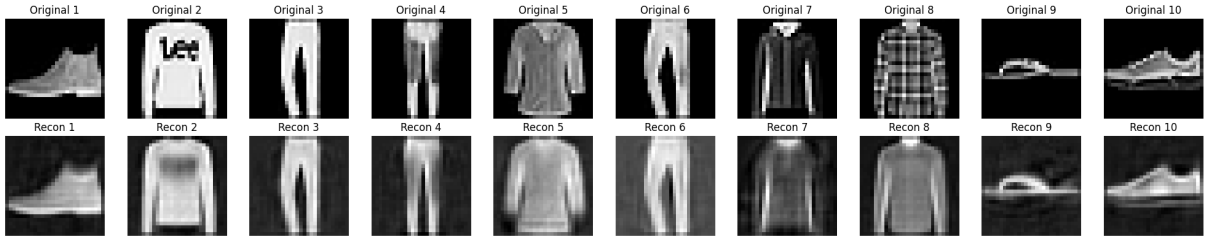


Figure 2: 32 dimension reconstruction

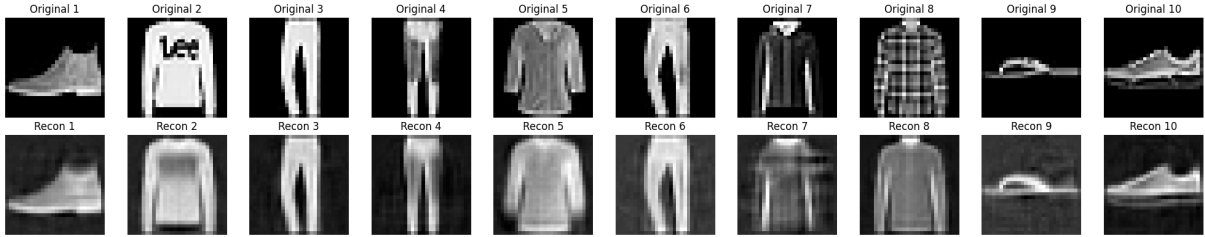


Figure 3: 64 dimension reconstruction

1.3 Interpolation between Classes

Interpolation was performed in latent space between a Trouser and an Ankle boot using the 64-dimensional model. The transition from trouser to ankle boot was almost smooth, but the midpoint image was blurry and showed properties of neither class clearly.

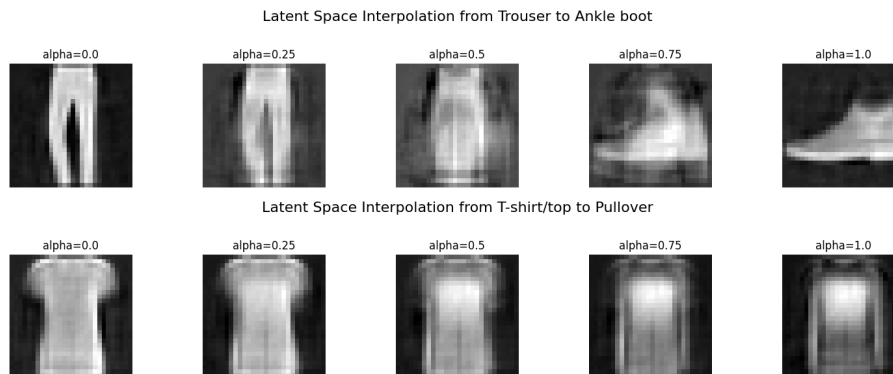


Figure 4: Interpolation

2 Representation Behaviour

2.1 What the Small Latent Model Loses

With only 2 latent dimensions, the model must compress 784 pixel values into 2 numbers. The major losses include:

- Logos, texts, etc on the shirt are lost.
- Sleeves are poorly reconstructed.
- Strap and buckle details on sandals and ankle boots are lost.

- The model tends to produce something which is recognisable but not sharp.

This extra capacity means the model is “memorising” more about individual training images rather than learning a general representation. From a feature-learning standpoint, a 64-dimensional space may offer no improvement over 32 because both already encode irrelevant variation.

2.2 When Reconstruction Looks Good But Feels Wrong

A reconstruction can have a low MSE score while still being semantically poor. This happens in two main situations:

Trivial memorisation. If a model with a large latent space has enough capacity, it can store near-copies of training images without learning structure. The reconstruction metric improves, but the latent codes do not encode anything interpretable or transferable.

Blurry correctness. The 2-dimensional model produces soft, averaged reconstructions. These have higher MSE, yet a human viewer might still recognise the class. On the other hand, blurry but similar reconstruction can hide the fact that all class-specific detail has been lost. The key question is not whether the pixel values look right, but whether the latent code captures the properties that distinguish one class from another.

3 Latent Space Analysis

3.1 Class Grouping and Separability

Visually confusable categories (T-shirt, Shirt, Pullover, Coat) form an overlapping cluster in the centre of the plot, which accurately reflects their visual ambiguity. More distinct categories (Bag, Trouser, Sandal) are well separated. For the 32 and 64 dimensional models, PCA was applied to reduce the latent codes to 2D for scatter plot visualisation. In both cases, visually similar categories cluster together:

3.2 Semantic Meaning in the Latent Space

The latent space does reflect semantic structure. The model did not explicitly see class labels during training, yet related items ended up near each other purely from reconstruction pressure. This suggests the model discovered that sharing latent codes for visually similar inputs lowers reconstruction error.

However, the representation is approximate. We can improve the model by using convolutional layers instead of fully connected ones, or a VAE.

4 Interpolation Analysis

4.1 Smoothness of Transitions

Transitions between semantically similar classes are smoother than transitions between dissimilar ones. When interpolating between Trouser and Ankle boot, the midpoint image is an incoherent blur because the two classes occupy very different regions of latent space.



Figure 5: Latent space analysis

The decoder has not been trained on any example from the middle of that path, so it produces an average that belongs to no real class.

4.2 Where Interpolation Breaks

- The two endpoints are semantically distant (e.g. Bag and Trouser), producing mid-points that look like faded noise.
- The latent dimension is small (2D), leaving no room for a meaningful path between classes; the path must pass through regions the encoder never mapped.
- The decoder is a simple MLP; without convolutional structure, it cannot cleanly reconstruct spatial transitions.